Sahaj Raj Malla

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Research Interests Multimodal agent orchestration, reinforcement learning, and compute-efficient decision-making.

Education

B.Sc. in Computational Mathematics Nov 2022 – Feb 2027 (expected)
Kathmandu University, Dhulikhel, Nepal · Expected degree conferral: Jun 2027
Undergraduate thesis: *When to Call Whom: The Value of Diagnostic Feedback in Budgeted AI Orchestration*
With a fixed budget, when should an agent review a failed attempt before trying another method?
Supervisors: Dr. Samir Shrestha and Dr. Saraswati Acharya

Selected Publications

Peer-Reviewed Publications

Q1 = top quartile in the journal's subject category (SCImago Journal Rank, 2025).

1. **S. R. Malla**. Heuristic optimization of AI agents for Dhumbal. *Scientific Reports* Q1 (Springer Nature), 16, 33849, Aug 2026.
   Compared AI strategies for the Dhumbal card game; an aggressive heuristic agent reached an 88.9% win rate.
2. **S. R. Malla**. MallaNet residual branch merge convolutional neural network with homogeneous filter capsules for Devanagari character recognition. *Scientific Reports* Q1 (Springer Nature), 16, 1175, Dec 2025.
   99.71% test accuracy on the Devanagari Handwritten Character Dataset, 56.41% fewer parameters than prior SOTA; introduces optimized residual blocks and Homogeneous Filter Capsule layers for multiscale feature extraction.
3. **S. R. Malla**. Combining rather than selecting multi-horizon forecasting of an agricultural commodity price index. *Scientific Reports* Q1 (Springer Nature), Nov 2026.
   Developed multi-horizon forecasting for a new agricultural commodity price index; direct change-target forecasting reduced errors by 33–53%, while shrinkage forecast combination improved accuracy by 1.3–11.2%.
4. **S. R. Malla**. A landmark-based addressing framework for urban navigation using geospatial clustering and pathfinding algorithm. *Kathmandu University Journal of Science, Engineering and Technology*, 19(1), Mar 2025.
   In simulation, 37.7% lower average path length and 22.9% lower travel time vs. current systems.
5. B. Kafle, A. Shrestha, **S. R. Malla**, S. Jaiswal. Evaluating XGBoost Algorithm for Nepali Stock Price Prediction. *Int. Conf. on Technologies for Computer, Electrical, Electronics & Communication (ICT-CEEL 2023)*, Oct 2023.
   XGBoost regression over 241,023 daily observations for 204 NEPSE-listed companies (2013–2023).

Preprints & Manuscripts Under Review

6. **S. R. Malla**. Devanagari Digit Recognition using Quantum Machine Learning. *arXiv:2506.09069*, Jun 2025.
   Hybrid quantum-classical architecture for Devanagari digit recognition (CNN + simulated 10-qubit variational circuit); 99.80% test accuracy, 0.9980 average per-class F1.
7. **S. R. Malla**. ArthaVyu: A Stacking Ensemble Framework for Multi-Source Real Estate Price Prediction in Nepal. *Research Square*, Jun 2026.
   Stacking ensemble framework for multi-source real estate price prediction in Nepal; unifies 24,673 property listings from seven platforms, reaching test R^2 0.7439 and MedAPE 26.91%.
8. **S. R. Malla**, S. Kayastha, R. Suwal, H. C. Bhandari, R. Adhikari. XGBoost Forecasting of NEPSE Index Log Returns with Walk Forward Validation. *arXiv:2601.08896*, Jan 2026.
   Walk-forward validation framework for time-series forecasting of NEPSE index log returns with XGBoost.
9. P. Panth, **S. R. Malla**. Secure, Verifiable, and Scalable Multi-Client Data Sharing via Consensus-Based Privacy-Preserving Data Distribution. *arXiv:2601.00418*, Jan 2026.
   Consensus-driven privacy-preserving framework for secure, verifiable multi-client data sharing.
10. **S. R. Malla**. External Demand, Domestic Monetary Conditions, and Remittance Dynamics in Nepal. *arXiv:2605.19401*, May 2026.
    ARDL-ECM model of remittances as a share of Nepal's GDP (1993–2024) using PCA-derived monetary indices.

Professional Experience

Founder Nov 2023 – Present
HamroNiwas, Banepa, Nepal

- Built a rental & marketplace platform for Nepali students, connecting 4,500+ users and reducing search time by 90%.

Data Engineer Consultant May 2025 – Jun 2025
YoungInnovations Pvt. Ltd., Lalitpur, Nepal

- Built ELT pipelines with Apache Spark for large-scale dataset processing; automated workflows with Apache Airflow.

Data Scientist Sep 2021 – Jan 2023
Naamche, Inc. (now reAlpha Nepal), Lalitpur, Nepal

- Scraped & analyzed 1M+ real estate listings (Florida, Texas, California); produced investment-decision reports, speeding report computation roughly threefold.

Data Engineer Trainee Mar 2021 – Sep 2021
Devfinity Nepal, Kathmandu, Nepal

- Built a Django-based data factory processing 1M+ job records daily, deployed on Azure.

Selected Projects

Touchstone: Multi-Agent Orchestration with Routing and Critic Review Nov 2024 – Present
Kathmandu University project, supervised by Dr. Harish Chandra Bhandari
LangGraph system routing queries to specialist agents and retrieval, with a critique agent reviewing responses.

GreatAPI Jun 2022 – Present
Personal project, open-source Python web framework
Full-stack FastAPI framework for AI; self-built and maintained.

SahajMails Nov 2025 – Present
Personal project, open-source Python library
AI-personalized bulk-email tool for outreach, newsletters, and campaigns; secure, cost-free email automation.

Patrakar: Nepali News Classification Aug 2022 – Sep 2022
Associated with IT MEET, open-source machine learning model
DistilBERT sequence-classification model for 9 Nepali news categories; 95.475% test accuracy.

Honors & Awards

Winner, Infinity Codewave Hackathon, Kathmandu University	May 2026
Winner, Infinity Data Visualization Hackathon, Kathmandu University	May 2026
Winner, Computer Engineering Model Demonstration, Kantipur Engineering College	Dec 2024
Winner, KU Hackfest (OSM Category), Kathmandu University Computer Club	Oct 2023
Winner, Seeds for the Future Nepal, Huawei Technologies Co., Ltd.	Sep 2023
Winner, Infinity Codewave, Department of Mathematics, Kathmandu University	Aug 2023
Winner, AI Competition (Text Challenge), IT MEET, Kathmandu University	Aug 2022
Winner, AI Competition (Data Visualization), IT MEET, Kathmandu University	Aug 2022
Winner, Youth Innovation Challenge, Social Changemakers and Innovators	Aug 2022

Conference Talks & Panels

Conference Presentation, Landmark-Based Addressing Oct 2024
Kathmandu University, Dhulikhel, Nepal

Panelist, National Youth Scientific Conference Jun 2024
National Youth Council Nepal, Kathmandu, Nepal

Guest Speaker, Career in AI/ML Dec 2023
CSIT Association of Nepal, Pokhara, Nepal

Conference Presentation, Nepali Stock Price Prediction Oct 2023
ICT-CEEL 2023, Bhaktapur, Nepal

Teaching & Volunteering

Teaching Assistant, Annual Nepal AI School Dec 2024 – Jan 2025
NAAMII, Lalitpur, Nepal

Website Manager & Executive Member May 2023 – May 2024
Kathmandu University Robotics Club, Dhulikhel, Nepal

Lead Quantum Computing Researcher Mar 2021 – Mar 2023
OneQuantum, Munich, Germany

Data Scientist & Software Developer Mar 2021 – Mar 2022
Robotics Association of Nepal, Kathmandu, Nepal

Technical Skills

Agents & RL:	Policy training, policy evaluation, offline RL, Gymnasium, MuJoCo
ML Frameworks:	PyTorch, JAX, Hugging Face Transformers, scikit-learn
LLM Systems:	LangGraph, RAG, LoRA fine-tuning, mixed-precision training
Efficient ML:	Parameter-efficient model design, model quantization, inference profiling
Software Engineering:	Python, C++, TypeScript, FastAPI, Django, Next.js
Cloud & Deployment:	AWS, Azure, Linux, Docker
Databases & Search:	PostgreSQL, MongoDB, Redis, FAISS
Data Engineering:	Apache Spark, Apache Airflow
Applied Mathematics:	Numerical optimization, statistical modeling, graph algorithms
Quantum ML:	PennyLane, Qiskit

Selected Training & Fellowships

Aspire Leaders Program, Aspire Institute	Jan 2025 – Mar 2025
Annual Nepal AI School, NAAMII	May 2023, Dec 2024
AI Fellowship, Fusemachines Nepal	Jan 2023 – Jan 2024
Womanium Quantum, Womanium & QWorld	Jul 2023 – Aug 2023
Data Science Fellowship with DataCamp, Code for Nepal	Dec 2021 – Aug 2023
Introduction to Quantum Computing, The Coding School & IBM	Sep 2021 – Apr 2022

Press & Media

AI Agent for the Dhumbal Card Game, *Techpana* (in Nepali)	Sep 2026
Making Devanagari Handwriting Accessible: The Story of MallaNet, *Business 360°*	Feb 2026
AI Model to Recognize Handwritten Devanagari Script, *Techpana* (in Nepali)	Dec 2025
South Asia's 1st Quantum AI for Devanagari Digits, *ICT Frame*	Jun 2025
Paths of Possibility Shown by Sahaj, *Terai Online* (in Nepali)	Apr 2025
A Journey with "Seeds for the Future 2023", *ICT Frame*	Nov 2023

Languages

English (IELTS overall 7.5), Nepali, Hindi, Newari